```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df=sns.load_dataset("titanic")
```

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

```
df["age"]=df["age"].fillna(df["age"].mean())
df.drop(["deck","alive"],axis=1,inplace=True)
df["embark_town"]=df["embark_town"].fillna(df["embark_town"].mode()[0])
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
embark_town	0
alone	0

dtype: int64

```
df=pd.get_dummies(df,drop_first=True)
```

```
x=df.drop("survived",axis=1)
y=df["survived"]
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
log_model=LogisticRegression()
log_model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (sta
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
▾ LogisticRegression
LogisticRegression()
```

```
y_pred=log_model.predict(x_test)
```

```
y_prob=log_model.predict_proba(x_test)
```

```
from sklearn.metrics import accuracy_score
accuracy=accuracy_score(y_test,y_pred)
print(accuracy)
```

```
0.8100558659217877
```

```
from sklearn.metrics import confusion_matrix
confusion=confusion_matrix(y_test,y_pred)
print(confusion)
```

```
[[91 14]
 [20 54]]
```

```
from sklearn.metrics import classification_report
report=classification_report(y_test,y_pred)
print(report)
```

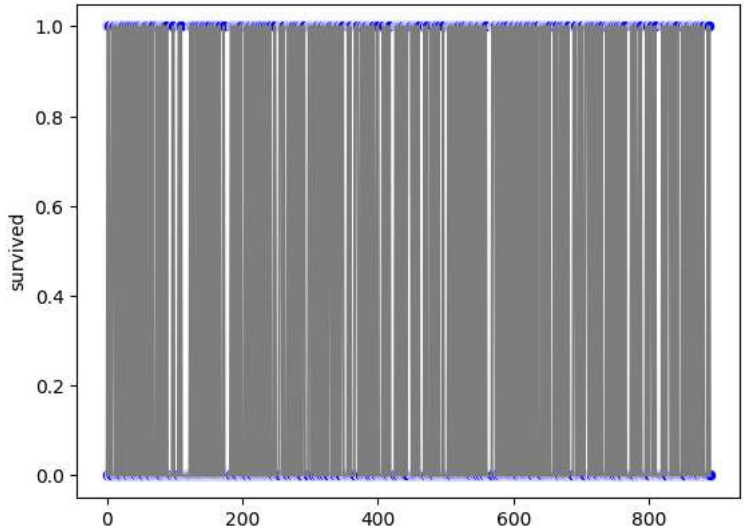
```
              precision    recall  f1-score   support

           0       0.82      0.87      0.84       105
           1       0.79      0.73      0.76        74

    accuracy                           0.81       179
   macro avg       0.81      0.80      0.80       179
weighted avg       0.81      0.81      0.81       179
```

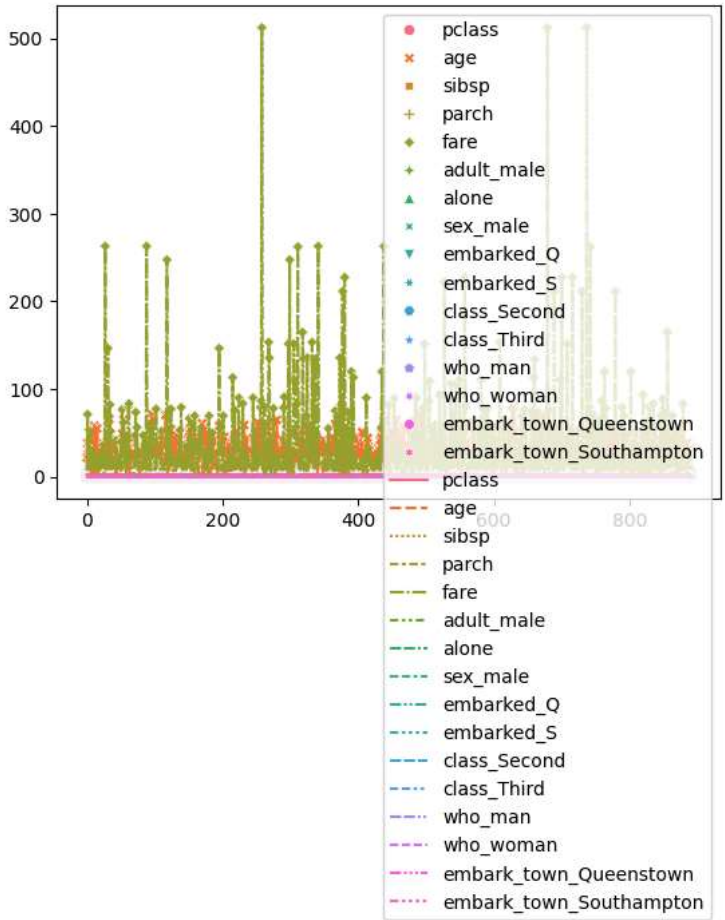
```
sns.scatterplot(y,color="blue")
sns.lineplot(y,color="gray")
```

```
<Axes: ylabel='survived'>
```



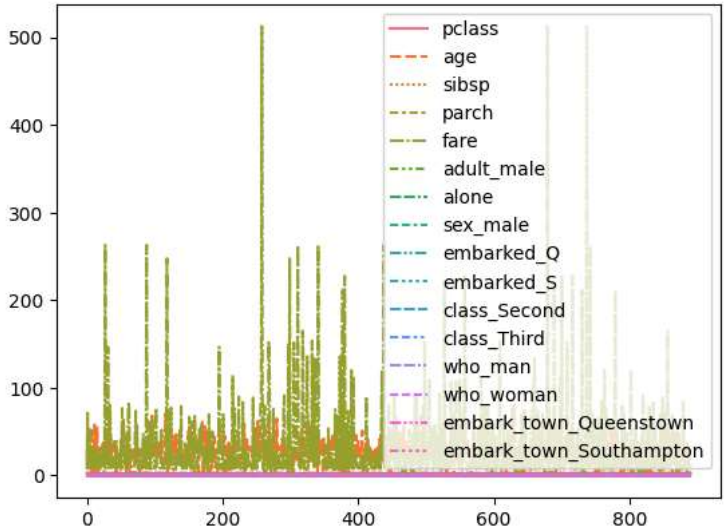

```
sns.scatterplot(x,color="green")
sns.lineplot(x ,color="yellow")
```

```
<Axes: >
```




```
sns.lineplot(x)
```

```
<Axes: >
```



Start coding or generate with AI.